



AI CHATBOT SYSTEM

Virtual Internship Project 1: Rule-Based Chatbot

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Internship Program: DecodeLabs Internship (Week 1)

Mission & Objectives

The Mission

To develop a robust, rule-based AI assistant capable of simulating human-like interaction through keyword-matching and logical conditional branching.

Core Objectives

Implement real-time user data handling, execute Python-based logic for responses, and ensure a seamless conversational experience for the user.

| Technology Stack



Python 3.x

The core language utilized for its simplicity and powerful string handling capabilities.



Datetime Library

Used to fetch and display current system date and time for utility-based queries.



Rule Logic

A comprehensive If-Elif-Else architecture designed for precise keyword recognition.

Interactive Key Features

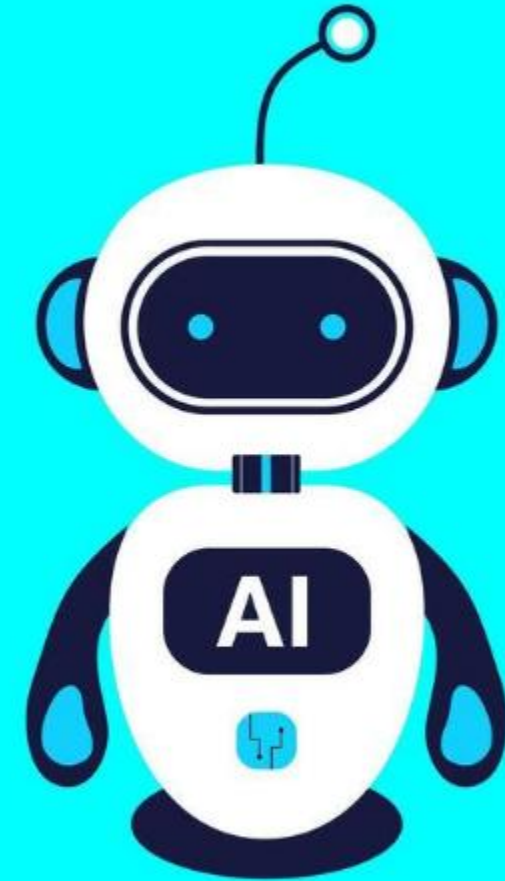
- 👤 **User Greetings:** Intelligent responses to various standard greetings like 'Hi', 'Hello', or 'Hey'.
- 🧠 **Memory Simulation:** Ability to store and repeat user-provided info like Name, Age, and Hobbies.
- 😊 **Entertainment:** Built-in joke generation and funny responses to keep the user engaged.



| System Logic

Logical Branching

The chatbot operates on a sequential logic path. It normalizes user input to lowercase to prevent case-sensitivity issues, ensuring 100% keyword match accuracy. By using string replacement methods, the bot extracts user variables (like Name or Age) directly from conversational sentences.



| The Conversation Flow



Input Reception: Bot listens for user text via the `input()` function within a continuous loop.



Normalization: Input is converted to lowercase for consistent comparison against the rule-base.



Keyword Mapping: The system scans the input for predefined substrings (e.g., "time", "joke", "name").



Dynamic Response: Bot provides either a static response or processes data (like time) to return dynamic content.

Live Terminal Output

DecodeBot AI Terminal - v1.0

Bot: Hello! How can I help you today?

You: my name is Salman

Bot: Nice to meet you Salman!

You: tell me a joke

Bot: Why did the computer go to school? Because it wanted to become smarter CPU 🤪

You: bye

Bot: Goodbye! Take care 😊



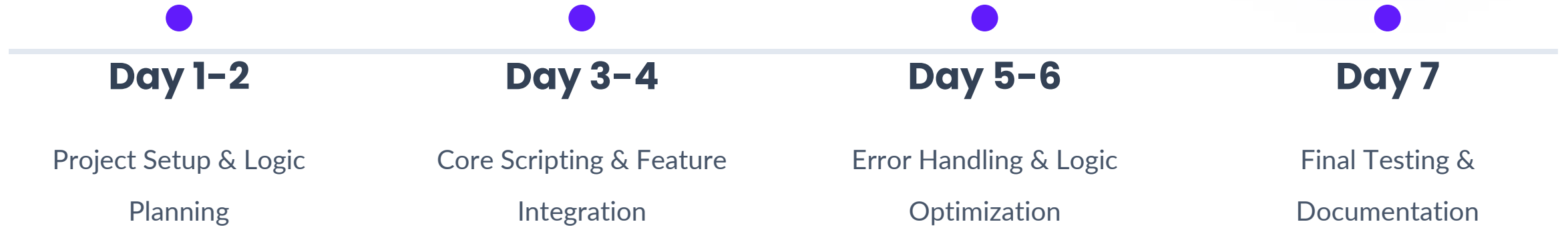
Stable Script Execution

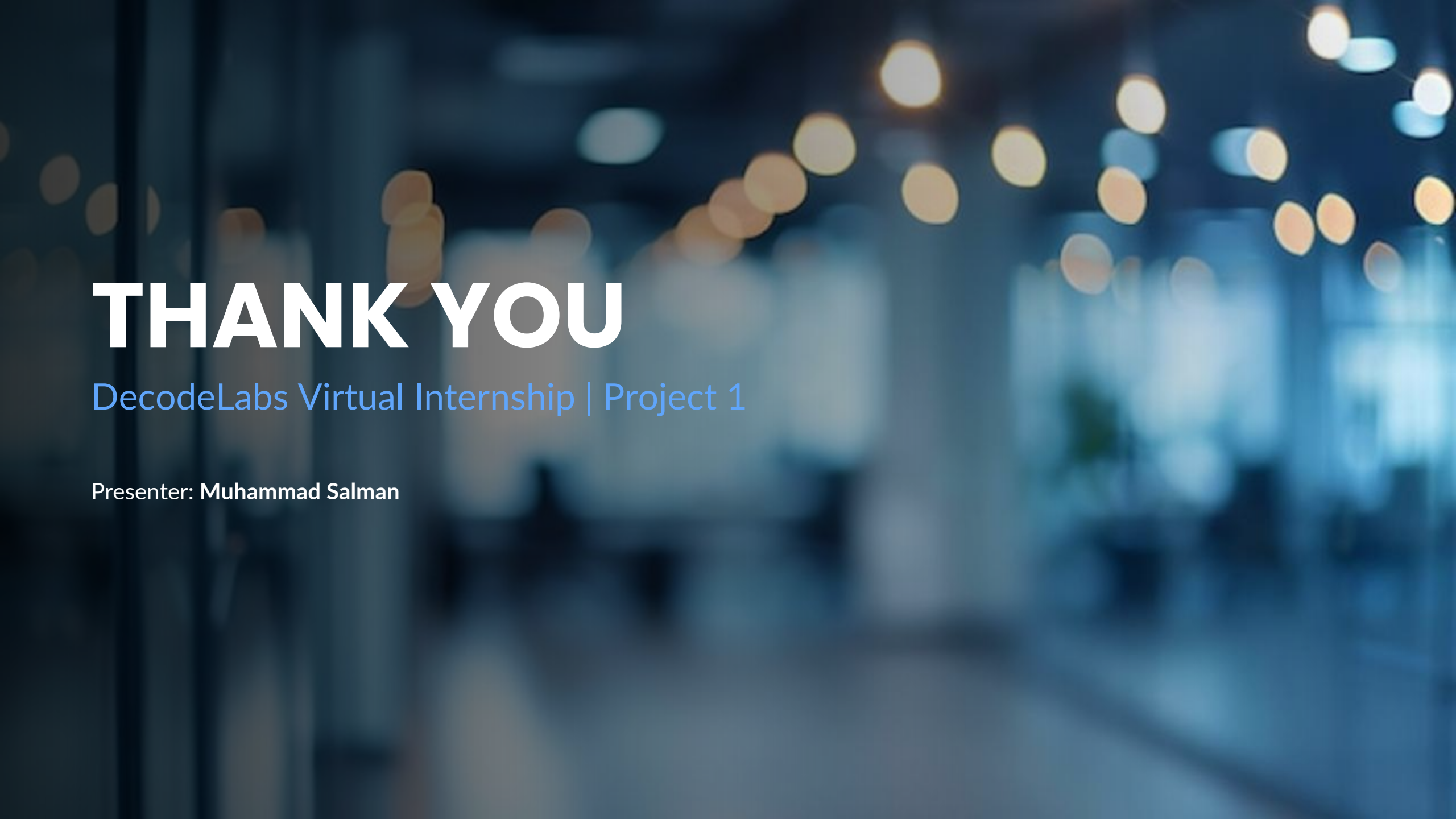
The console output demonstrates the script's ability to handle exit commands and personalization accurately.

Logic Keyword Dictionary

User Keyword	Bot Response Category	Dynamic Value Used
hi / hello / hey	Standard Greeting	None (Static)
my name is ...	Identity Storage	Variable: name
time / date	System Utility	datetime.now()
joke / funny	Entertainment	None (Static)
bye / exit / quit	Termination	Exit Script

Development Lifecycle (Week 1)





THANK YOU

DecodeLabs Virtual Internship | Project 1

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